

미적14주차 과제

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비판정법

$$1. \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{\frac{x^{n+1}}{(n+1)^5} \cdot \frac{5}{x}}{\frac{x^n}{n^5}} \right| = \lim_{n \rightarrow \infty} \frac{5x \cdot n^5}{(n+1)^5 \cdot 5} = \frac{1}{5} |x| < 1$$

$$R=5$$

교대수판정법

$$x = -5 \quad \sum_{n=1}^{\infty} \frac{(-5)^n}{n^5} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n^5} \quad \text{정확한}$$

$$x = 5 \quad \sum_{n=1}^{\infty} \frac{1}{n^5} \quad \text{收斂} \quad -5 \leq x \leq 5$$

비판정법

$$2. \frac{1}{4x} \rightarrow \frac{-\frac{1}{4}}{1-\frac{x}{4}} \rightarrow \sum_{n=0}^{\infty} \frac{1}{4} \left(\frac{x}{4}\right)^n = \sum_{n=0}^{\infty} \frac{1}{4^{n+1}} x^n \quad \frac{|x|}{4} < 1 \quad R=4$$

$$-\sum_{n=0}^{\infty} \frac{1}{4^{n+1}} \cdot \frac{1}{n!} x^{n+1} + C \quad f(0) = \ln 4 \quad \left(\ln 4 - \sum_{n=0}^{\infty} \frac{1}{4^{n+1}} \cdot \frac{1}{n!} x^{n+1} \right)$$

3.